

kdpcover: LaTeX Class for Amazon Book Covers*

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1 Introduction

This class produces a book cover suitable for Kindle Direct Publishing (KDP). The author employs it for a number of his books. A LaTeX document conforming to this class takes the following form:




```
\documentclass[pages=200,6x9]{kdpcover}
\begin{document}
\putSpine{kdpcover, a LaTeX class for KDP books}
\putVolume{1}
\putVersion{0.7.1}
\putPrice{\$0.00}
\putCopyright{\the\year}{Yegor Bugayenko}
\putTitle{kdpcover}
\putAuthor{Yegor Bugayenko}
\putTLDR{This book recounts
  an interesting story!}
\putPicture{cactus.pdf}
\end{document}
```

The problem addressed by this class is the necessity of adjusting the size of the cover PDF in accordance with the number of pages within the book—the more voluminous the book, the broader its spine must be. The class performs the requisite calculations on the fly by means of the `qpdf` tool, which must therefore be installed. The `pdflatex` processor must furthermore be invoked with the `--shell-escape` option, so that LaTeX may execute `qpdf`.

*The sources are on GitHub at yegor256/kdpcover

2 Class Options

`pages` By means of `pages`, the total number of pages within the book may be specified:

```
\documentclass[pages=120]{kdpcover}
\begin{document}
\end{document}
```

`pdf` The `pages` option is best avoided in favour of letting the class compute the size of the book from the PDF file (by means of the `qpdf` command-line tool, which must be installed):

```
\documentclass[pdf=book.pdf]{kdpcover}
\begin{document}
\end{document}
```

`6x9` This option sets the page dimensions to 6x9 inches. A custom size may likewise be defined as follows:

```
\documentclass[6x9]{kdpcover}
\setlength\kdpxsize{8.1in}
\setlength\kdpysize{12.7in}
\begin{document}
\end{document}
```

`7x10` The `7x10` and `8x10` options set the page dimensions to 7x10 and 8x10 inches
`8x10` respectively; `6x9` is the default.

`barless` To eliminate the black bar at the centre of the page, the `barless` option suffices:

```
\documentclass[barless]{kdpcover}
\begin{document}
\end{document}
```

3 Commands

`\putSpine` The `\putSpine` command places a vertical black spine line containing two white elements: the text at the top and the logo at the bottom. The optional argument names the file with the logo, `kdpcover-signature` by default. The text should be sufficiently short to avoid contact with the logo. It is sound practice to record the volume number at the end of the text whenever more than one volume exists, e.g. "my book, vol. 1". The use of small caps within the text alone is recommended.

`\putVolume` This command renders the volume figure. It is deliberately rendered at large size to ensure conspicuousness. Only volumes from 1 to 4 are supported, since the class ships the figures as `kdpcover-vol-1.pdf` through `kdpcover-vol-4.pdf`.

`\putVersion` This command prints the version and additionally appends the date of the latest Git commit in the current directory. Should the directory not constitute a Git repository, the date is omitted.

`\putPrice` This command prints the price, the inclusion of which is, in the author's view, sound practice.

`\putCopyright` This command prints a brief copyright notice, comprising the year and the name of the author.

`\putTitle` This command prints the title of the book.

`\putAuthor` This command prints the author.

`\putTLDR` This command prints the TL;DR paragraph, which summarises the central message of the entire book.

`\putPicture` This command prints the front-cover picture. The file `cactus.pdf` may be employed.

`\putBack` This command prints the text on the back of the book; the author generally recommends omitting it and leaving the back empty and white.

4 Implementation

First, we load our parent class:

```
\LoadClass{article}
```

Then, we define a few internal commands:

```
\makeatletter
\newcount\kdp@pages
  \kdp@pages=100
\newlength\kdpxsize
  \setlength\kdpxsize{6in}
\newlength\kdpysize
  \setlength\kdpysize{9in}
\makeatother
```

Then, we include `iexec` for being able to execute `qpdf` and `git`:

```
\RequirePackage{iexec}
```

Then, we parse class options with the help of the `pgfopts` package:

```
\RequirePackage{pgfopts}
\makeatletter
\pgfkeys{
  /kdp/.cd,
  barless/.store in = \kdp@barless,
  pdf/.code = {
    \IfFileExists{#1}
      {}
      {\ClassError{kdpcover}{The "#1" file is absent}{}}%
    \iexec[trace,quiet,stdout=kdpcover-pages-count.txt]
      {qpdf --show-npages --no-warn #1 | tr -d '[[:space:]]'}
    \openin1=kdpcover-pages-count.txt
    \read1 to \temp
    \closein1
    \kdp@pages=\temp
  },
  6x9/.code = {
    \setlength\kdpxsize{6in}
    \setlength\kdpysize{9in}
```

```
  },
  7x10/.code = {
    \setlength\kdpxsize{7in}
    \setlength\kdpysize{10in}
  },
  8x10/.code = {
    \setlength\kdpxsize{8in}
    \setlength\kdpysize{10in}
  },
  pages/.code = {
    \kdp@pages=#1
  }
}
\ProcessPgfPackageOptions{/kdp}
\makeatother
```

Then, we include a few useful packages:

```
\RequirePackage{anyfontsize}
\RequirePackage{tikz}
\RequirePackage{microtype}
\RequirePackage{xcolor}
\RequirePackage{graphicx}
\RequirePackage{calc}
```

Then, we set spacing using `setspace`:

```
\RequirePackage{setspace}
  \setstretch{1.2}
```

Then, we declare the lengths of the page. The height is `\kdpysize` plus 0.125 inches of "bleed" on top and on the bottom, see this. The width is twice `\kdpxsize` plus 0.125 inches of "bleed" on each side, plus the "spine," whose width depends on the number of pages in the book.

```
\makeatletter
\newlength\kdp@height
\newlength\kdp@width
\makeatother
```

Then, right before the document begins, we calculate the lengths, set the size of the page, using `geometry`, and split the page into a 16 by 16 grid of `textpos`, which must happen after the size is known:

```
\RequirePackage{geometry}
\makeatletter
\AddToHook{begindocument/before}{%
  \setlength\kdp@height{0.125in + \kdpysize + 0.125in}%
  \setlength\kdp@width{
    0.125in + \kdpxsize
    + 0.125in + \kdpxsize
    + 0.0025in * \kdp@pages}%
  \geometry{
    paperwidth=\kdp@width,
    paperheight=\kdp@height,
    left=0pt,
    right=0pt,
    top=0pt,
    bottom=0pt
```

```
  }%
  \TPGrid{16}{16}%
}
\makeatother
```

Then, we load `textpos` with absolute coordinates:

```
\PassOptionsToPackage{absolute}{textpos}
\RequirePackage{textpos}
```

`\putSpine` Then, we define `\putSpine`:

```
\makeatletter
\newcommand\putSpine[2][kdpcover-signature]{%
  \ifdefined\kdp@barless\else%
    \begin{textblock}{2.4}[0.5,0](8,0)%
      \begin{tikzpicture}%
        \node [rectangle, inner sep=0em, fill=black,
        minimum width=2.4\TPHorizModule,
        minimum height=16\TPVertModule] at (0,0) {};
      \end{tikzpicture}%
    \end{textblock}%
  \fi%
  \begin{textblock}{1}[0.5,0](8,2)%
    \begin{tikzpicture}%
      \node [color=white, inner sep=0cm, outer sep=0cm,
        rotate=270, minimum height=\TPHorizModule] at (0,0) {
        \Large #2%
      };
    \end{tikzpicture}%
  \end{textblock}%
  \begin{textblock}{2.4}[0.5,1](8,14)%
    \centerline{\includegraphics[width=0.32in]{#1}}%
  \end{textblock}%
}
\makeatother
```

`\putPicture` Then, we define `\putPicture`:

```
\newcommand\putPicture[1]{%
  \begin{textblock}{4}(10,2)%
    \includegraphics[width=\textwidth]{#1}
  \end{textblock}%
}
```

`\putVolume` Then, we define `\putVolume`:

```
\newcommand\putVolume[1]{%
  \begin{textblock}{2}[1,1](15,14)%
    \raggedleft%
    \includegraphics[height=0.4in]{kdpcover-vol-#1}%
  \end{textblock}%
}
```

`\putPrice` Then, we define `\putPrice`:

```
\newcommand\putPrice[1]{%
  \begin{textblock}{4}[0,1](1,2)%
    \small #1%
```

```
  \end{textblock}%
}
```

\putBack Then, we define \putBack:

```
\newcommand\putBack[1]{%
  \begin{textblock}{5}[0,0](1,3)%
    \small #1%
  \end{textblock}%
}
```

\putTitle Then, we define \putTitle:

```
\newcommand\putTitle[1]{%
  \begin{textblock}{5}(10,9)%
    \fontsize{32}{32}\selectfont #1%
  \end{textblock}%
}
```

\putAuthor Then, we define \putAuthor:

```
\newcommand\putAuthor[1]{%
  \begin{textblock}{4}(10,10)%
    \large by #1%
  \end{textblock}%
}
```

\putTLDR Then, we define \putTLDR:

```
\newcommand\putTLDR[1]{%
  \begin{textblock}{5}(10,11)%
    TL;DR #1%
  \end{textblock}%
}
```

\putVersion Then, we define \putVersion. The date is printed when the current directory is a Git repository, which means that .git is either a directory with HEAD inside or, in a worktree or a submodule, a plain file. Since kpathsea refuses to open directories, .git is checked twice:

```
\makeatletter
\newcommand\kdp@gitdate{%
  \quad\iexec{git log -n 1 --pretty='format:\%ad' --date='format:\%e-\%b-\%Y'}%
}
\newcommand\putVersion[1]{%
  \begin{textblock}{4}[0,1](10,14)%
    #1%
    \IfFileExists{.git/HEAD}
      {\kdp@gitdate}
      {\IfFileExists{.git}{\kdp@gitdate}{}}%
  \end{textblock}%
}
\makeatother
```

\putCopyright Then, we define \putCopyright:

```
\newcommand\putCopyright[2]{%
  \begin{textblock}{4}[0,1](1,14)%
    \small (c)~%
```

```
    #1 #2%
  \end{textblock}%
}
```

Finally, a few layout instructions:

```
\AtBeginDocument{%
  \ttfamily%
  \raggedright%
  \setlength\parindent{0pt}%
  \setlength\parskip{0pt}%
  \interfootnotelinepenalty=10000%
}
\endinput
```

Change History

v0.2.0

General: Initial version 3

v0.5.0

General: The `docshots` package is used to render the documentation. 1

The base class is now `article`. . 3

v0.6.1

General: If the PDF file is absent, an error is printed. 3

v0.6.2

General: The `textpos` grid is computed from the cover size instead of the letter paper. . . . 4

`\putVersion`: The Git date is printed when `.git` is a directory, not only when it is a file. 6

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